

Standard Library

Python provides several such useful values (constants), classes and functions.

This collection of predefined utilities is referred as the **Python Standard Library**

All these functionalities are organized into different modules.

- In Python context, any file containing a Python code is called a **module**
- These modules are further organized into folders known as **packages**

Different modules are:

1. collections
2. random
3. datetime
4. math and many more..

Working with Standard Library

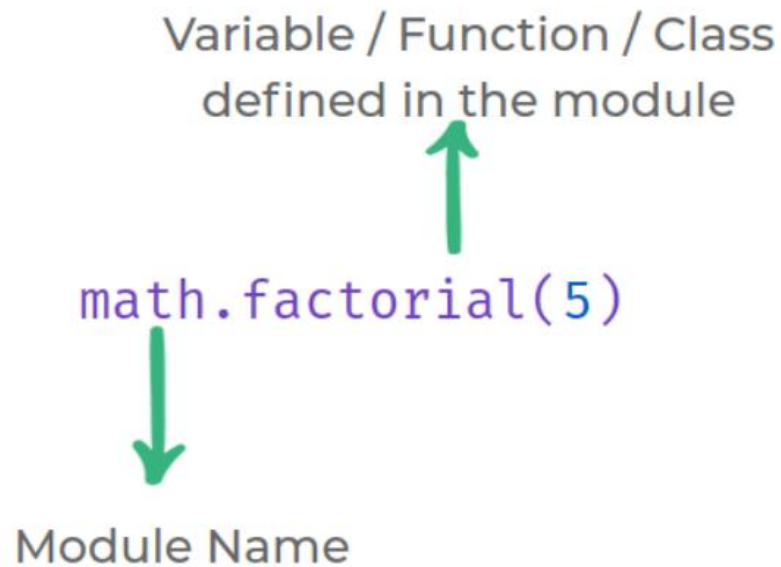
To use a functionality defined in a module we need to import that module in our program.

```
1 import module_name
```

PYTHON

Math Module

math module provides us to access some common *math functions* and *constants*.



Code

PYTHON

```
1 import math
2 print(math.factorial(5))
3 print(math.pi)
```

Output

```
120
3.141592653589793
```

Importing module

Importing a module and giving it a new name (aliasing)

Code

PYTHON

```
1 import math as m1
2 print(m1.factorial(5))
```

Output

120

Importing from a Module

We can import just a specific definition from a module.

Code

PYTHON

```
1 from math import factorial
2 print(factorial(5))
```

Output

120

Aliasing Imports

We can also import a specific definition from a module and alias it

Code

PYTHON

```
1 from math import factorial as fact
2 print(fact(5))
```

Output

120

Random module

Randomness is useful in whenever uncertainty is required.

For example: Rolling a dice, flipping a coin, etc.,

`random` module provides us utilities to create randomness.



Randint

`randint()` is a function in random module which returns a random integer in the given interval.

Code

PYTHON

```
1 import random
2 random_integer = random.randint(1, 10)
3 print(random_integer)
```

Output

8

Choice

`choice()` is a function in random module which returns a random element from the sequence.

Code

PYTHON

```
1 import random
2 random_ele = random.choice(["A","B","C"])
3 print(random_ele)
```

Output

B

To know more about **Python Standard Library**, go through the authentic python documentation

- <https://docs.python.org/3/library/>

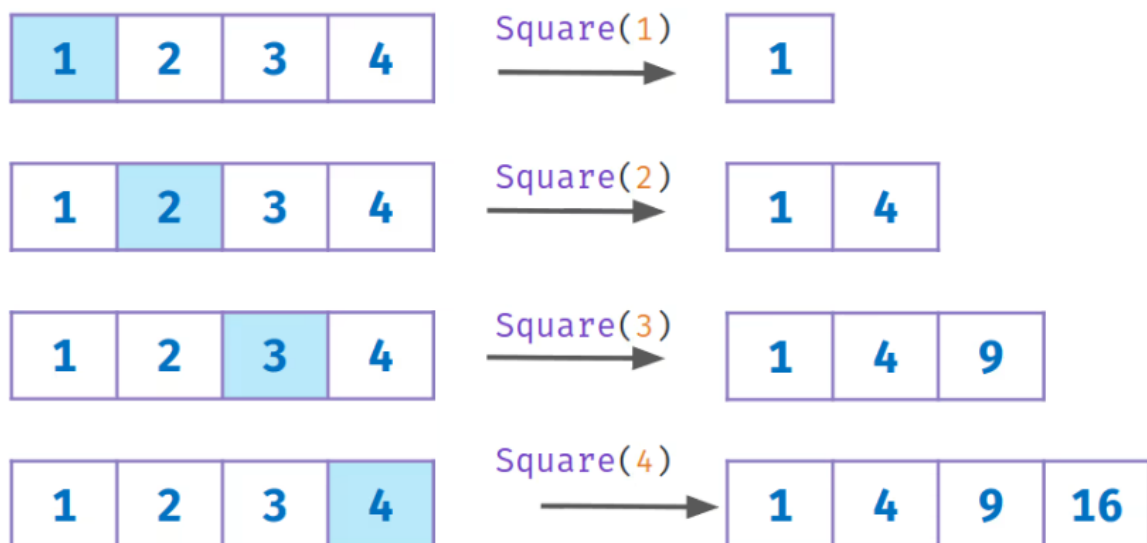
Map, Filter and Reduce

We worked with different sequences (list, tuples, etc.)

To simplify working with sequences we can use `map()` , `filter()` and `reduce()` functions.

Map

`map()` applies a given function to each item of a sequence (list, tuple etc.) and returns a sequence of the results.



Example - 1

Code

PYTHON

```
1 def square(n):  
2     return n * n  
3 numbers = [1, 2, 3, 4]  
4 result = map(square, numbers)  
5 numbers_square = list(result)  
6 print(numbers_square)
```

Output

```
[1, 4, 9, 16]
```

Example - 2

Code

PYTHON

```
1 numbers = list(map(int, input().split()))  
2 print(numbers)
```

Input

```
1 2 3 4
```

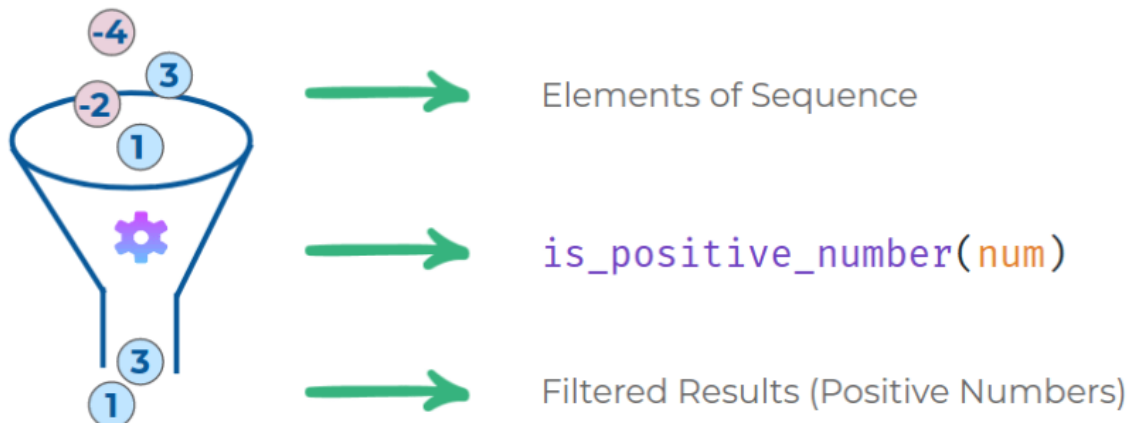
Output

```
[1, 2, 3, 4]
```

Filter

`filter()` method filters the elements of a given sequence based on the result of given function.

The function should return True/False



Code

```
1 def is_positive_number(num):  
2     return num > 0  
3  
4 list_a = [1, -2, 3, -4]  
5 positive_nums = filter(is_positive_number, list_a)  
6 print(list(positive_nums))
```

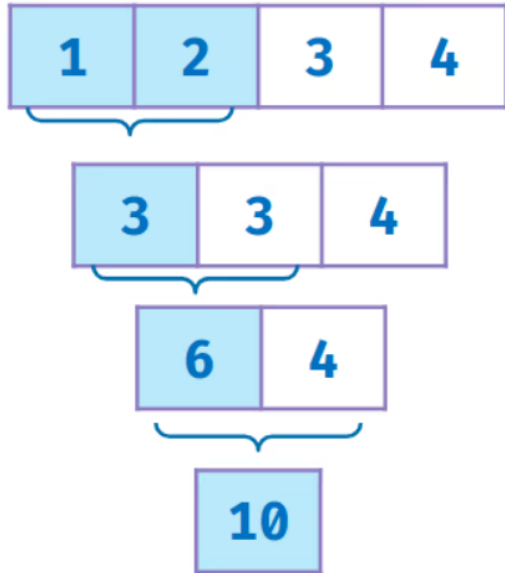
PYTHON

Output

```
[1, 3]
```


Reduce

`reduce()` function is defined in the `functools` module.



Code

```
1 from functools import reduce
2
3 def sum_of_num(a, b):
4     return a+b
5
6 list_a = [1, 2, 3, 4]
7 sum_of_list = reduce(sum_of_num, list_a)
8 print(sum_of_list)
```

PYTHON

Output

10